

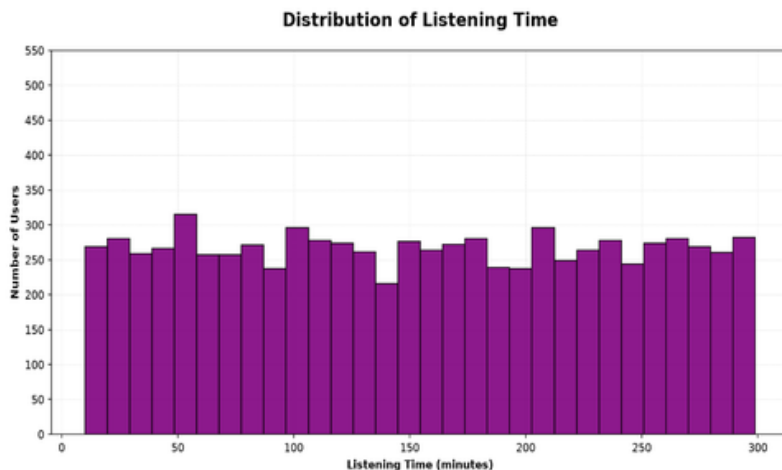
OBJECTIVE

The objective of this analysis is to examine user listening behavior and identity factors associated with customer churn. Python libraries such as Pandas, Numpy , Matplotlib, and Seaborn were used to clean, analyze, and visualize the dataset. The findings can help understand user engagement and support strategies to improve customer retention.

TOOLS USED

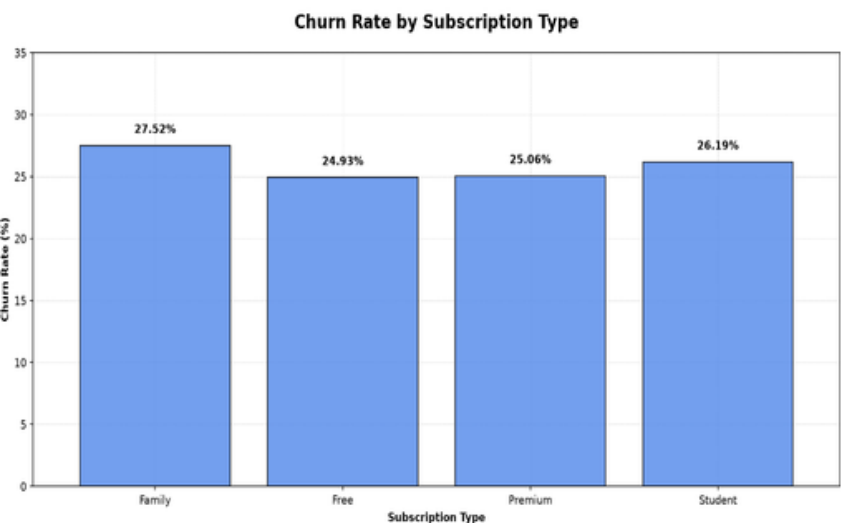
- VS CODE
- Python
- NumPy
- Matplotlib
- Seaborn

Chart 1 : Distribution of Listening Time (Histogram)



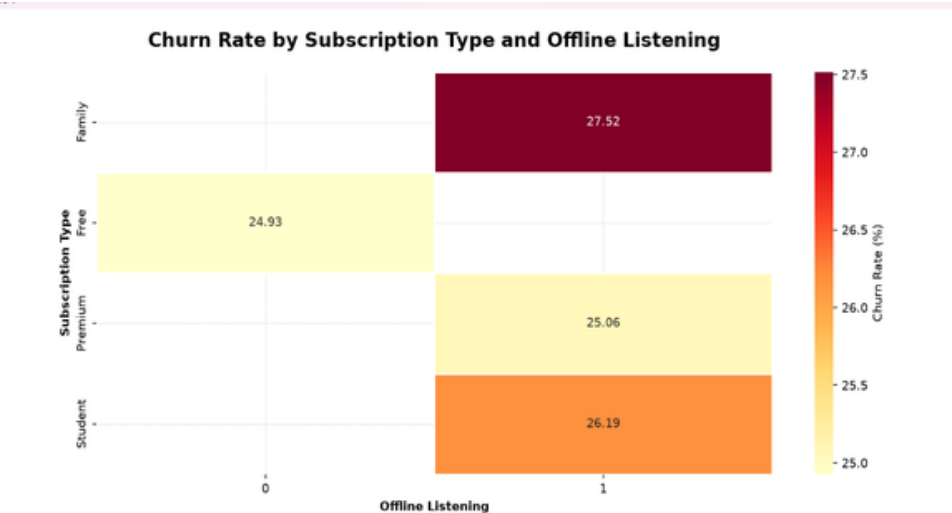
- **Reason :** A Histogram was chosen because it is the most suitable chart for understanding the distribution of a continuous variable such as listening time. It helps identify where most users fall, whether the data is evenly distributed, and if there are users with exceptionally high or low listening times.
- **Insight :** The listening time data shows a **highly uniform distribution** across the 10 to 300 minute range, with each bar holding a similar value of users (avg. 250-300 users per tier). This means we don't have a huge group of super-users or a huge group of people who barely open the app.
- **How it helps in business :** Because the usage is so even, raw listening time by itself doesn't tell us who is satisfied or who is about to leave. It tells the business that we need to mix this data with actual churn status to see if the users with lower listening time are the ones canceling.

Chart 2 : Churn Rate by Subscription Type (Bar Chart)



- **Reason :** I used a bar chart because it is the easiest way to compare different categories side-by-side. I wanted to see if users on a specific plan cancel more often than others.
- **Insight :** The churn rates are similar across all plans (about 25-27.5%). However, Family and Student plans are showing slightly higher churn rates (about 27.5%), while premium and free tiers are slightly lower at 25%.
- **How it helps in Business :** The marketing team can focus retention efforts on Family and Student accounts (as these plans show slightly higher churn rates) through targeted offers and service improvements.

Chart 3 : Churn Rate by Subscription Type and Offline Listening (Heat Map)



- **Reason** : A heat map is suitable for looking at two different factors at the same time using colors. Here, I wanted to see how combining plan types with "Offline Listening" (0 for No, 1 for Yes) is associated with the churn rate.
- **Insight** : It shows a major trend: Free users only churn when they don't use offline listening (24.93%). On the other hand, paid users (Family, Premium, Student) only churn when they do use offline listening, with Family plans hitting the highest peak at 27.52%.
- **How it helps the business**: This gives the product team something specific to fix. Since paid users who use offline listening are the ones leaving. The company should check on the offline listening experience for paid users to find out what issue is causing the churn.

How the Charts Are Related

I chose these three charts because each one helps answer a different question about user behavior and customer churn. **The histogram** was selected to understand how users spend their listening time and to get an overall view of user engagement. Since listening time is a continuous variable, a histogram is the most suitable chart to show its distribution.

The bar chart was chosen to compare churn rates across different subscription plans. It clearly shows which subscription types have relatively higher customer churn, making it easier to identify user groups that may need more attention.

The heatmap was selected to compare churn rates based on both subscription type and offline listening. By combining these two factors, it becomes easier to identify which groups of users have higher or lower churn rates and where the company should focus its efforts to keep the customers.

These charts are related because they build the analysis step by step. The histogram provides an overview of user activity, the bar chart compares churn across subscription plans, and the heatmap combines subscription type with offline listening to identify churn patterns. Together, they provide useful insights for better business decisions.